Two-Root Book-Edge Leaves and Their Cones: A Geometric-Mean Graphon Bound

Abstract

For integers $m, a, b \geq 1$, let $B_{m,a,b}$ be a book of m triangles sharing one edge, with a pendant leaves at one end of that edge and b pendant leaves at the other. We prove, for every graphon W of edge density $p \geq 1/2$,

$$t(B_{m,a,b}, W) \geq p^{a+b+1}(2p-1)^m = \Phi_{B_{m,a,b}}(p).$$

The proof replaces a previously unsuccessful signed-moment argument by a geometric-mean compression under the edge-biased probability measure. A normalized-edge Cauchy–Schwarz estimate gives the exact mean needed for Jensen's inequality. We then prove the corresponding cone theorem for $K_1 \vee B_{m,a,b}$ on the full interval $p \geq 2/3$. In the catalogue on at most six vertices, the newly positive graphs are Atlas 35, 93, 112, and 180.

1 Graphons and the book-edge families

Let (Ω, μ) be an arbitrary probability space. A graphon is a symmetric measurable function $W\colon \Omega^2 \to [0,1]$. For every finite simple graph F, its ordinary, non-injective homomorphism density is

$$t(F, W) = \int_{\Omega^{V(F)}} \prod_{uv \in E(F)} W(x_u, x_v) \prod_{u \in V(F)} \mathrm{d}\mu(x_u).$$

All integrands below are measurable and bounded, so Tonelli–Fubini applies. Write

$$p = t(K_2, W), \qquad d(x) = \int_\Omega W(x,y)\,\mathrm{d}\mu(y).$$

Fix an edge rs. For integers $m, a, b \geq 1$, define $B_{m,a,b}$ by adjoining pairwise distinct page vertices $t_1, \ldots, t_m$, each adjacent precisely to r and s, then adjoining leaves $u_1, \ldots, u_a$ adjacent only to r and leaves $v_1, \ldots, v_b$ adjacent only to s. Thus the triangles rst_i form an m-page book with spine rs, and both ends of the spine receive a nonempty pendant-leaf set. Interchanging a and b gives an isomorphic graph. The earlier two-root triangle-leaf family is the special case $m = 1$.

The second family consists of the cones

$$C_{m,a,b} := K_1 \vee B_{m,a,b}.$$

Thus the new cone vertex is adjacent to every vertex of $B_{m,a,b}$, including all of its pendant leaves.

Put $n = a + b$. Once the spine has been properly coloured, every leaf has $x - 1$ independent colour choices, while every page vertex has $x - 2$ choices after the spine has been coloured. Hence

$$\chi_{B_{m,a,b}}(x) = x(x-1)^{n+1}(x-2)^m, \qquad \chi(B_{m,a,b}) = 3. \tag{1}$$

For the cone, colouring the cone vertex first gives

$$\chi_{C_{m,a,b}}(x) = x\chi_{B_{m,a,b}}(x-1) = x(x-1)(x-2)^{n+1}(x-3)^m, \qquad \chi(C_{m,a,b}) = 4. \tag{2}$$

For $q = 1 - p > 0$, direct substitution in the definition of the chromatic target gives

$$\Phi_{B_{m,a,b}}(p) = q^{n+m+2}\frac{1}{q}\left(\frac{p}{q}\right)^{n+1}\left(\frac{2p-1}{q}\right)^m = p^{n+1}(2p-1)^m, \tag{3}$$

$$\begin{aligned}\Phi_{C_{m,a,b}}(p) &= q^{n+m+3}\frac{1}{q}\frac{p}{q}\left(\frac{2p-1}{q}\right)^{n+1}\left(\frac{3p-2}{q}\right)^m \\ &= p(2p-1)^{n+1}(3p-2)^m.\end{aligned} \tag{4}$$

Both right-hand sides are polynomials and therefore give the continued values at $p = 1$, namely 1.

2 The normalized-edge geometric-mean estimate

The following lemma is the new analytic ingredient.

Lemma 2.1 (Geometric mean of endpoint degrees)**.** *Suppose $p > 0$, and put*

$$Z(x,y) := \sqrt{d(x)d(y)}.$$

Then

$$\int_{\Omega^2} W(x,y)Z(x,y)\,\mathrm{d}\mu(x)\,\mathrm{d}\mu(y) \geq p^2. \tag{5}$$

Equivalently, under the edge-biased probability measure

$$\mathrm{d}\rho(x,y) = \frac{W(x,y)}{p}\,\mathrm{d}\mu(x)\,\mathrm{d}\mu(y),$$

one has $\mathbb{E}_\rho Z \geq p$.

Proof. If $d(x) = 0$, then $W(x,y) = 0$ for almost every y, because $W \geq 0$ and $d(x) = \int W(x,y)\,\mathrm{d}\mu(y)$. Thus all quotients below may be defined as zero when one of their degree denominators vanishes.

Cauchy–Schwarz first gives

$$\begin{aligned}\int_{\Omega^2}\frac{W(x,y)}{Z(x,y)}\,\mathrm{d}\mu(x)\,\mathrm{d}\mu(y) &= \int_{\Omega^2}\sqrt{\frac{W(x,y)}{d(x)}}\sqrt{\frac{W(x,y)}{d(y)}}\,\mathrm{d}\mu(x)\,\mathrm{d}\mu(y) \\ &\leq \left(\int_{\Omega^2}\frac{W(x,y)}{d(x)}\,\mathrm{d}\mu^2\right)^{1/2}\left(\int_{\Omega^2}\frac{W(x,y)}{d(y)}\,\mathrm{d}\mu^2\right)^{1/2} \leq 1.\end{aligned}$$

Indeed, the first integral in the product equals $\mu\{x : d(x) > 0\} \leq 1$, and the second is identical after exchanging x and y.

Apply Cauchy–Schwarz a second time to $\sqrt{WZ}$ and $\sqrt{W/Z}$. The zero-degree observation justifies the pointwise factorization on the support of W, and hence

$$p^2 = \left(\int_{\Omega^2} W\,\mathrm{d}\mu^2\right)^2 \leq \left(\int_{\Omega^2} WZ\,\mathrm{d}\mu^2\right)\left(\int_{\Omega^2}\frac{W}{Z}\,\mathrm{d}\mu^2\right) \leq \int_{\Omega^2} WZ\,\mathrm{d}\mu^2.$$

This is (5). □

Lemma 2.2 (Convex book truncation)**.** *For all integers $n \geq 0$ and $m \geq 1$, the function*

$$f_{n,m}(z) := z^n \max\{2z-1, 0\}^m, \qquad 0 \leq z \leq 1,$$

where $z^0 = 1$, is convex and nondecreasing.

Proof. The function vanishes on $[0, 1/2]$. On $[1/2, 1]$ it is a product of n copies of the nonnegative increasing affine factor z and m copies of the nonnegative increasing affine factor $2z-1$. Its first derivative is a sum of products of nonnegative factors and positive slopes; its second derivative is a sum over ordered pairs of differentiated factors, again with only nonnegative factors and positive slopes. It is therefore nondecreasing and convex on $[1/2, 1]$. Its value at $1/2$ is zero and its right derivative there is nonnegative (positive when $m = 1$, zero when $m \geq 2$), whereas the left derivative is zero. The one-sided derivatives thus increase across the joining point, proving the assertion on all of $[0, 1]$. □

3 Triangle books with leaves at both spine ends

Theorem 3.1 (Two-root book-edge leaf theorem)**.** *For every $m, a, b \geq 1$ and every graphon W of edge density $p \in [1/2, 1]$,*

$$\boxed{t(B_{m,a,b}, W) \geq p^{a+b+1}(2p-1)^m = \Phi_{B_{m,a,b}}(p).}$$

This is the complete interval required by $\chi(B_{m,a,b}) = 3$.

Proof. The case $p = 1$ is treated separately below, so assume first that $1/2 \leq p < 1$. Integrate out all leaves, and then independently integrate out the m page vertices. With

$$c(x, y) := \int_\Omega W(x, z)W(y, z)\,\mathrm{d}\mu(z),$$

one obtains the exact identity

$$t(B_{m,a,b}, W) = \int_{\Omega^2} W(x, y)d(x)^a d(y)^b c(x, y)^m\,\mathrm{d}\mu^2. \tag{6}$$

For numbers $A, B \in [0, 1]$, one has $AB \geq A + B - 1$. Applying this to $W(x, z)$ and $W(y, z)$ and integrating in z gives, almost everywhere,

$$c(x, y) \geq d(x) + d(y) - 1.$$

Also $c \geq 0$, and arithmetic–geometric mean gives

$$c(x, y) \geq \max\{d(x) + d(y) - 1, 0\} \geq \max\{2\sqrt{d(x)d(y)} - 1, 0\}.$$

The factors W and c^m are symmetric in x, y. Consequently, (6) is unchanged if a and b are exchanged, and it equals the average of the two resulting integrals. Since all factors are nonnegative, (3) and the arithmetic–geometric mean inequality imply, with $n = a + b$ and $Z = \sqrt{d(x)d(y)}$,

$$\begin{aligned} t(B_{m,a,b}, W) &\geq \int_{\Omega^2} W(x, y)\frac{d(x)^a d(y)^b + d(x)^b d(y)^a}{2} \max\{2Z-1, 0\}^m\,\mathrm{d}\mu^2 \\ &\geq \int_{\Omega^2} W(x, y)(d(x)d(y))^{(a+b)/2} \max\{2Z-1, 0\}^m\,\mathrm{d}\mu^2 \\ &= p\,\mathbb{E}_\rho f_{n,m}(Z). \end{aligned}$$

Lemma 2.2, Jensen's inequality, Lemma 2.1, and monotonicity of $f_{n,m}$ now give

$$t(B_{m,a,b}, W) \geq p f_{n,m}(\mathbb{E}_\rho Z) \geq p f_{n,m}(p) = p^{n+1}(2p-1)^m.$$

The last expression is the target in (3).

If $p = 1$, then $0 \leq W \leq 1$ and $\int W = 1$ force $W = 1$ almost everywhere. Hence $t(B_{m,a,b}, W) = 1$, which equals the polynomially continued target. □

At the threshold $p = 1/2$, the target is zero; no division by this factor or by a triangle density occurs. The proof above remains valid because the edge-biased measure is well-defined. Alternatively, the threshold statement follows immediately from nonnegativity of homomorphism densities.

4 A self-contained conditional cone lemma

We reproduce the accepted link argument in the exact form needed for the new family. This makes clear why the next result is a conditional theorem, not a general universal-vertex closure.

Define the rooted triangle density

$$\tau(x) = \int_{\Omega^2} W(x,y)W(x,z)W(y,z)\,\mathrm{d}\mu(y)\,\mathrm{d}\mu(z).$$

Lemma 4.1 (Weighted rooted-triangle inequality)**.** *If $p > 0$, then for every integer $h \geq 0$,*

$$\int_\Omega d(x)^h \tau(x)\,\mathrm{d}\mu(x) \geq \frac{2p-1}{p} \int_\Omega d(x)^{h+2}\,\mathrm{d}\mu(x).$$

Proof. Put $M_j = \int d^j$, let T_W be the integral operator with kernel W, and set

$$I_h = \int d^h \tau, \qquad J_h = \left\langle d^h, T_W d \right\rangle.$$

The inequality $AB \geq A + B - 1$, applied to $W(x,y), W(x,z)$ and multiplied by $d(x)^h W(y,z)$, gives

$$I_h \geq 2J_h - pM_h. \tag{7}$$

Put $U = 1 - W$, $u = T_U \mathbf{1} = 1 - d$, and $q = 1 - p$. Since

$$T_W d = d - q\mathbf{1} + T_U u,$$

we have

$$J_h = M_{h+1} - qM_h + \left\langle d^h, T_U u \right\rangle. \tag{8}$$

By symmetry of U, twice the difference

$$\left\langle d^h, T_U u \right\rangle - \int d^h u^2$$

equals

$$\int_{\Omega^2} U(x,y)\big(u(y) - u(x)\big)\big(d(x)^h - d(y)^h\big)\,\mathrm{d}\mu^2 \geq 0,$$

because $d = 1 - u$, so u and d^h are oppositely ordered. Substitution in (8) gives

$$J_h \geq M_{h+2} - M_{h+1} + pM_h.$$

Together with (7),

$$I_h \geq 2M_{h+2} - 2M_{h+1} + pM_h.$$

Finally, subtracting $(2p-1)M_{h+2}/p$ from the right-hand side leaves

$$\frac{1}{p}\int_\Omega d(x)^h (d(x)-p)^2 \,\mathrm{d}\mu(x) \geq 0.$$

□

For $d(x) > 0$, define the probability measure

$$\mathrm{d}\mu_x(y) = \frac{W(x,y)}{d(x)}\,\mathrm{d}\mu(y),$$

and regard W as a graphon W_x on (Ω, μ_x). Its edge density is $z_x = \tau(x)/d(x)^2$. Put $z_x = 0$ when $d(x) = 0$. The preceding lemma immediately yields, for every integer $N \geq 2$,

$$\int_\Omega d(x)^N z_x \,\mathrm{d}\mu(x) \geq \left(2 - \frac{1}{p}\right)\int_\Omega d(x)^N \,\mathrm{d}\mu(x). \tag{9}$$

Lemma 4.2 (Conditional cone lemma)**.** *Let F have $N \geq 2$ vertices, put $\phi(z) = \Phi_F(z)$, and suppose that for some $\alpha \in [0,1)$:*

1. *every graphon V of edge density $z \geq \alpha$ satisfies $t(F,V) \geq \phi(z)$;*
2. *$\phi \geq 0$, $\phi' \geq 0$, and $\phi'' \geq 0$ on $[\alpha, 1]$;*
3. *$\phi(\alpha) = 0$.*

Then every graphon W of edge density

$$p \geq \frac{1}{2-\alpha}$$

satisfies

$$t(K_1 \vee F, W) \geq \Phi_{K_1 \vee F}(p).$$

Proof. The endpoint $p = 1$ is immediate, so assume $p < 1$ and put $c = 2 - 1/p$. The density hypothesis is exactly $c \geq \alpha$, and $c \leq 1$. Convexity shows that every graphon V of arbitrary edge density $z \in [0,1]$ satisfies the global tangent bound

$$t(F,V) \geq \phi(c) + \phi'(c)(z-c).$$

Indeed, for $z \geq \alpha$ this follows from the assumed graphon bound and convexity. For $z < \alpha$, the tangent is nonpositive: it is at most $\phi(\alpha) = 0$ at α and has nonnegative slope. Nonnegativity of homomorphism densities handles this second range.

Conditioning on the image of the cone vertex gives the exact identity

$$t(K_1 \vee F, W) = \int_\Omega d(x)^N t(F, W_x)\,\mathrm{d}\mu(x).$$

There is no division issue on $\{d = 0\}$, because its contribution is zero. Let $M_N = \int d^N$. The tangent bound and (9) give

$$\begin{aligned} t(K_1 \vee F, W) &\geq M_N \phi(c) + \phi'(c)\left(\int d(x)^N z_x \,\mathrm{d}\mu(x) - cM_N\right) \\ &\geq M_N \phi(c) \geq p^N \phi(c), \end{aligned}$$

where the last step is Jensen and uses $\phi(c) \geq 0$.

For $q = 1 - p > 0$, one has $1 - c = q/p$ and $\chi_{K_1 \vee F}(x) = x\chi_F(x-1)$. Therefore

$$p^N \phi(c) = q^N \chi_F\left(\frac{p}{q}\right) = q^{N+1} \chi_{K_1 \vee F}\left(\frac{1}{q}\right) = \Phi_{K_1 \vee F}(p).$$

□

5 Cones over the two-root book family

Theorem 5.1 (Two-root book-edge leaf cones)**.** *For every $m, a, b \geq 1$ and every graphon W of edge density $p \in [2/3, 1]$,*

$$\boxed{t(C_{m,a,b}, W) \geq p(2p-1)^{a+b+1}(3p-2)^m = \Phi_{C_{m,a,b}}(p).}$$

This is the complete interval required by $\chi(C_{m,a,b}) = 4$.

Proof. By Theorem 3.1, the base $B_{m,a,b}$ has the graphon lower bound

$$\phi_{n,m}(z) = z^{n+1}(2z-1)^m, \qquad n = a + b,$$

throughout $z \geq 1/2$. On $[1/2, 1]$, this polynomial is a product of $n+1$ copies of the nonnegative increasing affine factor z and m copies of the nonnegative increasing affine factor $2z - 1$. Its first and second derivatives are sums of products of nonnegative factors and positive slopes. Thus $\phi_{n,m} \geq 0$, $\phi'_{n,m} \geq 0$, and $\phi''_{n,m} \geq 0$ there, while $\phi_{n,m}(1/2) = 0$.

Lemma 4.2, with $\alpha = 1/2$, applies for

$$p \geq \frac{1}{2 - 1/2} = \frac{2}{3}.$$

It proves the required graphon inequality. Formula (4) identifies the exact target. □

6 Endpoints, equality examples, and catalogue consequences

At $p = 1/2$, the base target vanishes, and the balanced complete bipartite graphon attains equality because $B_{m,a,b}$ contains a triangle. At $p = 2/3$, the cone target vanishes, and the balanced complete tripartite graphon attains equality because $C_{m,a,b}$ contains a K_4. At $p = 1$, the argument given above shows that $W = 1$ almost everywhere and both sides equal 1.

More generally, for an integer $k \geq 2$, the balanced complete k-partite graphon T_k has edge density $p = 1 - 1/k$ and satisfies

$$t(F, T_k) = \frac{\chi_F(k)}{k^{v(F)}} = \Phi_F(p).$$

This applies to $B_{m,a,b}$ for $k \geq 2$, and to $C_{m,a,b}$ for $k \geq 3$. No claim is made here that these are all equality cases.

Corollary 6.1 (New positive Atlas cases)**.** *The following formerly open catalogue rows satisfy the desired inequality throughout their complete required intervals:*

Atlas	*graph6*	v	e	χ	*structure*	$\Phi_H(p)$
35	`Dx_`	5	5	3	$B_{1,1,1}$	$p^3(2p-1)$
93	`EC{O`	6	6	3	$B_{1,1,2}$	$p^4(2p-1)$
112	`EhX_`	6	7	3	$B_{2,1,1}$	$p^3(2p-1)^2$
180	`EjtW`	6	10	4	$C_{1,1,1}$	$p(2p-1)^3(3p-2)$

Their chromatic polynomials are, respectively,

$$x(x-1)^3(x-2), \qquad x(x-1)^4(x-2), \qquad x(x-1)^3(x-2)^2, \qquad x(x-1)(x-2)^3(x-3).$$

Proof. For unambiguous identification, the NetworkX Atlas edge lists are

$$\begin{aligned} 35: &\quad \{01, 02, 04, 12, 23\},\\ 93: &\quad \{03, 04, 14, 24, 34, 35\},\\ 112: &\quad \{01, 12, 14, 15, 23, 24, 25\},\\ 180: &\quad \{01, 04, 12, 13, 14, 15, 23, 34, 35, 45\}. \end{aligned}$$

In Atlas 35, 012 is a triangle with one leaf at 0 and one at 2. In Atlas 93, 034 is a triangle with one leaf at 3 and two at 4. In Atlas 112, the edge 12 is shared by the triangles 124 and 125, with one leaf at each end of 12. Thus Theorem 3.1 applies. In Atlas 180, vertex 1 is universal; deleting it leaves a copy of $B_{1,1,1}$ on the other five vertices, so Theorem 5.1 applies. The orders, sizes, chromatic polynomials, and targets follow from (1)–(4). □

7 Repair of the earlier obstruction and hidden-assumption audit

The earlier direct attempt for $B_{1,1,1}$ used $W(x,z)W(y,z) \geq W(x,z) + W(y,z) - 1$ and produced a difference of moments with a negative coefficient. Substituting a lower bound into that negative term reverses the useful direction, so that proof cannot be repaired by further algebra of the same two moments.

The proof here repairs precisely that step. It keeps the positive part of the common-neighbour estimate, symmetrizes the two rooted degree powers, and compresses them to $Z = \sqrt{d(x)d(y)}$. Lemma 2.1 then supplies the missing exact lower bound $\mathbb{E}_\rho Z \geq p$, and the convex function $f_{n,m}$ has no coefficient with the wrong sign.

Theorems 3.1 and 5.1 apply directly to arbitrary graphons on arbitrary probability spaces. Quotients on zero-degree sets are assigned zero only after proving that W vanishes there almost everywhere. Every integral is of a bounded nonnegative measurable function, apart from explicitly integrable normalized quotients. The proof uses no regularity, finite-step representation, finite-graph approximation, injective-copy count, minimum-degree condition, pointwise lower bound on d, or assumption that a conditioning event has positive mass.

The only standard analytic inputs are Tonelli–Fubini, arithmetic–geometric mean, Cauchy–Schwarz, and Jensen. The weighted link mechanism in Section 4 is also the conditional cone lemma accepted in `notes/six_verified_base_cones.pdf`; it was reproved here in full.

8 Exact limitations

The base theorem requires a book of triangles with one common spine edge, and every remaining vertex must be a pendant leaf at one of the two spine ends; both leaf sets must be nonempty in the catalogue predicate. It does not cover a leaf at a page vertex, a two-edge tail, a tree branching away from the book, or attachments to unrelated core graphs. The cone theorem covers only $K_1 \vee B_{m,a,b}$ and depends on the explicitly verified convex target $z^{a+b+1}(2z-1)^m$. It is not a general universal-vertex theorem and proves no arbitrary graph join.

No conclusion is asserted for graphs merely resembling these families, and no numerical experiment or finite sampling is used in either proof.

9 What the Lean formalization actually proves

The machine-checked development is `lean/Taeyoung/Methods/GeometricMean.lean` together with `lean/Taeyoung/Methods/TwoRoot/{Core,Rows}.lean`, ending at `satisfiesLowerBound_{35,93,112,180}`. It follows this note except at one place.

The departure: Lemma 2.2 is not formalized. This note establishes that $f_{n,m}(z) = z^n \max\{2z-1, 0\}^m$ is convex and nondecreasing on $[0,1]$, by comparing one-sided derivatives across $z = 1/2$, and then applies Jensen. The formalization never mentions convexity, derivatives, or `ConvexOn`. It instead proves the affine minorant that Jensen would have produced, directly:

$$f_{n,m}(p) + f'_{n,m}(p)(w-p) \le f_{n,m}(w) \qquad \text{for all } w \ge 0,\ p \ge 1/2, \tag{10}$$

as `TwoRoot.tangent_trunc`, where $f_{n,m}$ and $f'_{n,m}$ are the formal product and product-rule derivative of the root list $1^n 2^m$ in `Methods/AffineProduct.lean`. For $w \ge 1/2$ the truncation is inactive and (10) is the existing tangent lemma `affineProd_tangent` for products of nonnegative affine factors. For $w < 1/2$ the right-hand side is zero, while the left-hand side is at most its value at $w = 1/2$, because the slope $f'_{n,m}(p)$ is nonnegative; and that value is at most $f_{n,m}(1/2) = 0$ by the same tangent lemma. So the case split on the sign of $2w-1$ happens once, pointwise in a lemma about real numbers, and never inside an integral.

Integrating (10) against the edge-biased measure needs only that the slope is nonnegative and that $\mathbb{E}_\rho[Z] \ge p$, which is Lemma 2.1. The result is the same bound this note obtains from Jensen. The two routes agree in strength; the formal one avoids `ConvexOn`, one-sided derivatives, and the second-derivative argument in the proof of Lemma 2.2, none of which have a convenient Mathlib interface for a truncated product.

Everything else transcribes directly. The geometric-mean estimate is `GeometricMean.sq_le_integral_geoIn` proved by the two Cauchy–Schwarz steps of this note, carried out in $[0,\infty]$ so that the unbounded quotients W/d need no integrability side condition; the pointwise factorizations there hold almost everywhere, and the null set is handled by the same argument as in `notes/whiskering.pdf`. The two-vertex page bound $H_0 \ge d(x)+d(y)-1$ is `Link.le_pageOp_zero` and the arithmetic–geometric mean step is `GeometricMean.two_mul_geoMean_le`. The symmetrization for $a \ne b$, needed only for Atlas 93, is `MeasureTheory.integral_prod_swap` followed by $d(x)d(y)^2 + d(x)^2 d(y) = Z^2(d(x)+d(y)) \ge 2Z^3$. The cone (Atlas 180) uses the already-formalized conditional cone lemma rather than the reproved version in Section 4 of this note.